

Digital Logic

Day 3

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uarch.space/slides/day3.pdf

Counting

When counting, every next number has a new symbol:

0 → 1 → 2 → 3 → 4

But when we reach nine, we run out of symbols, so we start again from zero, with an extra place value.

8 → 9 → 10

The '1' in '10' represents the quantity ten.

Counting

We normally use base 10 for counting, because we have 10 symbols, 0-9.

Computers use base 2 instead, because there are only two symbols:   (1, 0).

Each place value is 2x the previous one.

Base 10:

12

$$1(10) + 2(1)$$

Base 2:

1100

$$1(8) + 1(4) + 0(2) + 0(1)$$

[Binary Demo](#)

now you try

1 Bit Decoder

The input for this level represents a 1 bit number, which can be 0 or 1.

A *decoder* takes this binary number and outputs **ON only** to the output of that number.

For example, if the input is 0, OUT 0 should be the only output that is on.



OUT 0



OUT 1

now you try

2 Bit Decoder

Every time a new bit is added,
the number of possible numbers
that can be represented
doubles.

2 bits can represent 0, 1, 2,
and 3!

Try to focus on one
output at a time, as if
you are making 4
different circuits.